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CHE 313 — Past Questions & Materials
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Institution: Federal University of Petroleum Resources, Effurun
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Levels: 300

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Paper 1 of 1 — CHE 313 Chemical Kinetics 2024/2025

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



COLLEGE OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF CHEMICAL ENGINEERING

FIRST SEMESTER EXAMINATION, 2024/2025 ACADEMIC SESSION

COURSE TITLE: Chemical Kinetics I

COURSE CODE: CHE 313 COURSE UNIT: 3 UNIT

TIME ALLOWED: 3 HOURS

INSTRUCTION: There are seven (7) Questions, answer any five. All questions carry equal mark. Start each question on a fresh page

1. (a) Define the following terms relating to chemical kinetics

- (i) Rate of reaction
- (ii) Non-elementary reaction
- (iii) Elementary reaction
- (iv) Molecularity
- (v) Order of reaction

b. Butadiene C_4H_6 , dimerizes in a Diels-Alder condensation to yield a substituted cyclohexane C_8H_{12} . Given the data on the 400K gas phase reaction below, show that the dimerization occurs as a second-order process, and find the rate constant.

Time (s)	Total Pressure (torr)
0	626
750	579
1500	545
2460	510
3425	485
4280	465
5140	450
6000	440
7500	425
9000	410
10500	405

Hint: 1 torr = 1/760 atm

CET/10/30/2022

2. (a) A rocket engine burns in a stoichiometric mixture of fuel (Liquid Hydrogen) in oxidant (liquid oxygen). The combustion chamber is cylindrical, 75cm long and 60 cm in diameter, and the combustion process produces 108kg/s of exhaust gases. If the combustion is complete, find the rate of reaction of H_2 and O_2

- b. The following values were obtained for the second-order rate constant of a biomolecular reaction at the temperature indicated

Temp, K	Rate constant ($dm^3 mol^{-1} sec^{-1}$)
285	1.07×10^{-2}
290	2.82×10^{-2}
298	0.126
306	0.525

- (i) Calculate the activation energy
(ii) Calculate the enthalpy of activation at 300K

3. (a) A reaction has the rate constant $k = 0.01 sec^{-1}$ for $C_{AO} = 20 mol/L$. What time is required for a 90% conversion in a constant volume batch reactor?



- (b) Calculate ε_A (expansion factor) for the following reaction conditions giving the reaction:



- (i) 50% A, 50% inert
(ii) 70% A, 30% inert
(iii) 30% A, 70% inert
(iv) 60% A, 40% inert

b. Find the rate equation to represent the data below using integral methods

Time (s)	0.00	2	4	6
CA (mol/L)	2.00	1.31	0.95	0.73

4 (a) An experimental rate measurement on the decomposition of A is made with a particular catalyst (see pertinent data listed below).

Data, for the spherical particle:

$d_p = 2.4 \text{ mm}$ or $L = R/3 = 0.4 \text{ mm} = 4.0 \times 10^{-4} \text{ m cat}$

$\phi_e = 5 \times 10^{-5} \text{ m}^3/\text{hr.m cat}$ (effective mass conductivity)

$k_{\text{eff}} = 1.6 \text{ kJ/hr.m cat.K}$ (effective thermal conductivity)

For the gas film surrounding the pellet (from correlations in the literature):

$h = 160 \text{ kJ/hr.m}^2 \text{ cat}$ (heat transfer coefficient)

$k_g = 300 \text{ m}^3/\text{hr.m}^2 \text{ cat}$ (mass transfer coefficient)

For the reaction:

$\Delta H = -160 \text{ kJ/mol A}$ (exothermic)

$C_{A,g} = 20 \text{ mol/m}^3$ (at 1 atm and 336°C)

$-r'_{A,obs} = 10^3 \text{ mol/hr.m}^3 \text{ cat}$

Assume that the reaction is first order.

(i) Is it likely that film resistance to mass transfer influences the rate? (2 marks)

(ii) Could this run have been made in the regime of strong pore diffusion? (2 marks)

(iii) Would you expect to have temperature variations within the pellet or across the gas film? (4 marks)

(b). As a Chemical Engineer discuss the heat effect during reaction, write down supporting equations. (6 marks)

5.(a) The catalytic reaction is a study in a plug flow reactor using various amount of catalyst and 20 liter/hr of pure A fed at 3.2 atm and 117°C . The concentration of A in the effluent stream is recorded for various runs as follows:

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Runs	1	2	3	4
Catalyst used wt. (kg)	0.02	0.04	0.08	0.16
C_{Aout} mol/liter.	0.074	0.06	0.044	0.029

Find the rate equation for this reaction using the integral method of analysis.

Given $C_{AO} = 0.1$ mol/liter, $F_{AO} = 2$ mol/liter, $\xi_A = 3$ (10 marks)

(b) Consider the catalytic reaction, use the rate equation given below to determine the amount of catalyst needed in a packed bed reactor (assume mixed flow) for 35% conversion of A to R for a feed of 2000 mol/hr of pure A at 3.2 atm and 117°C.

$A \rightarrow 4R$, $C_{AO} = 0.1$ mol/liter, $F_{AO} = 2$ mol/liter, $\xi_A = 3$ and $-r_A = 96C_A$ mol/kg.cat.hr.

(4 marks)

6. A single catalytic packed bed reactor is to be designed to treat 100 mol/s of reactant A and produce product R. Feed gas enters at 2.49 MPa and 300 K, the maximum allowable temperature is 900 K unless otherwise noted, the product stream is wanted at 300 K, and the thermodynamics and kinetics of the exothermic reaction are given, find attached graph. Prepare a sketch showing the details of the system you plan to use:

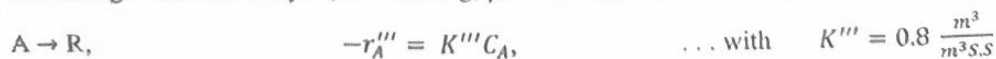
- The type of reactor: plug, recycle, or mixed (∞ recycle)
- The amount of catalyst needed
- The heat duty ahead of the reactor, at the reactor itself, and after the reactor
- The temperature of all flowing streams.

In this problem, assume that we are dealing with ideal gases.

$C_p = 40$ J/mol.K for all materials, and at all temperatures. This means that ΔH_r has the same value at all temperatures.

Design a single adiabatic packed bed system and work out a good design for 80% conversion of a feed consisting of 1 mol A and 7 mol inert. (14 marks)

7. First-Order Catalytic Reaction involving a Bubbling Fluidized bed. Reactant gas ($u_o = 0.3$ m/s, $v_o = 0.3\pi$ m³/s) passes upward through a 2-m diameter fluidized bed, $u_{mf} = 0.03$ m/s, $e_{mf} = 0.5$) containing 7 tons of catalyst ($W = 7000$ kg, $\rho_s = 2000$ kg/m³). Reaction proceeds as follows:



(a) Calculate the conversion of reactant. (8 marks)

(b) Find the proper mean concentration of A seen by the solids. (2 marks)

(c) If gas were made to flow downward through the solids we would have a packed bed.

Assuming plug flow of gas, find the conversion of reactant for this situation. (2 marks)

(d) Also sketch the reactor that represents this system. (2 marks)

Additional Data

$$C_{AO} = 100 \text{ mol/m}^3, \quad \mathcal{D} = 20 \times 10^{-6} \text{ m}^2/\text{s}, \quad \alpha = 0.33$$

Estimated bubble size in the bed: $d_b = 0.32 \text{ m}$.



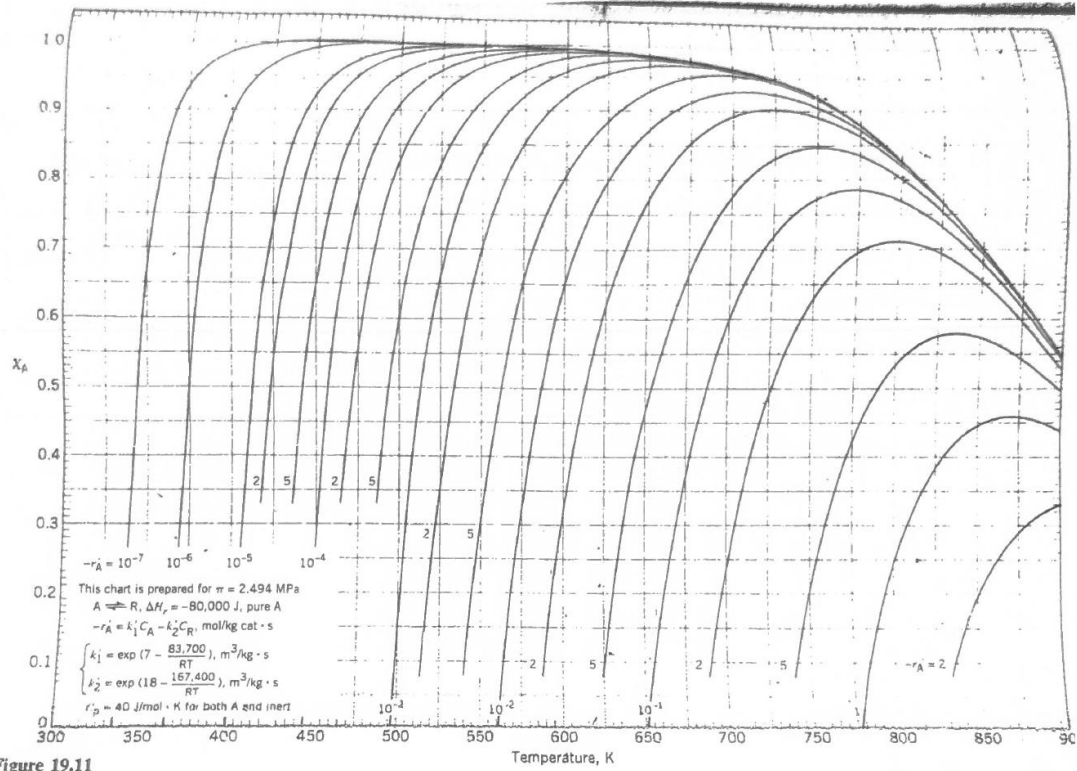


Figure 19.11